

Benford's Law for Detecting Contrast Enhancement

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Abstract—

Detection of contrast enhancement operations is a quite researched topic in the field of Digital Image Forensics as the forgers often use it to conceal the visual traces left by some malicious altering of the image, such as cut-and-paste. Benford's law is an important concept from the field of statistics and it has been found to be followed by a large number of natural phenomena. In this paper, a possibility of using Benford's law for the detection of contrast enhancement has been analysed and it is found that Benford's law based features show promising capability in differentiating between contrast enhanced images and unaltered images. Further, results demonstrate that it also withstands an existing anti-forensic technique used to hide the artifacts introduced by contrast enhancement operation.

Keywords—image tampering, contrast enhancement, anti-forensics.

I. INTRODUCTION

Nowadays, the availability of cheap image editing software tools have resulted in large number of images being modified with the intention to deceive someone. These circumstances have boosted the research in digital images forensics where the goal is to verify the authenticity of a digital image by either identifying its source of origin or reconstructing its history of processing. Researchers in this field have developed various techniques ([1], [2], [3], [4], etc.) to detect different kind of forgeries. The foundation of such techniques is based on the fact that image has characteristics which depend upon the components of the imaging devices and that these image characteristics are specific to the respective imaging device. Also, any post-processing applied to an image either creates distinct intrinsic fingerprints that can be identified or it disturbs the imaging device based image characteristics that can be detected for the purpose of verifying an image's authenticity. During the Life-cycle of an image, three kind of fingerprints are introduced into it [5] viz. acquisition, coding, and editing fingerprints.

Digital Image Forensics is a quite broad term but if we analyse the existing literature, it has been narrowed down to its specific forms such as *Passive*, *Active*, *Blind*, or *Non-Blind*. Depending upon the interference of forensic investigator into the image generation process, we can classify digital image forensics as Passive or Active. Similarly, based on the availability of prior knowledge of the original image we have Blind and Non-Blind image forensics.

Digital image anti-forensics has the goal of obstructing the forensic analyses of digital images. It works by concealing

the manipulation fingerprints or altering the acquisition fingerprints in order to mislead the analyst. A lot of work has been done some of which primarily focusses a particular editing operation such as in [6], the sudden peak and gap artifacts introduced by contrast enhancement are avoided by increasing the internal bit depth.

Benford's Law is an important concept in Statistics which deals with the distribution of each digit in numerical data. According to this law, each digit follows logarithmic law as the most significant digit in many naturally occurring data. It was first discovered by Simon Newcomb [7] in 1881 and later refined by Frank Benford [8] in 1938 stating that the frequency of appearance of each digit as the leading digit is given by,

$$p(d) = \log_{10}(1 + 1/d) \quad (1)$$

Out of the various fields, Benford's law is found to be applicable in image processing as well. Since the Discrete Cosine Transform (DCT) coefficients are found to follow the Benford's law [9], this law has been used to estimate the JPEG 2000 compression [10], multiple JPEG compression [11], JPEG steganography detection [12], etc. In [13], pre-processing filters are detected using six kind of features including the Benford's law based features. The rest of this paper is organised as follows. In section II, contrast enhancement operation and its characteristic fingerprints are discussed. In the next section, the proposed idea is given along with the details of the analysis done. The section IV presents the experimental results and finally the last section contains the conclusion drawn out of this work.

II. DETECTION OF CONTRAST ENHANCEMENT

Whenever the pixels of an image are concentrated in a narrow range, the image tends to have poor contrast such as *mostly dark*, *mostly bright*, or *mostly medium*. In order to enhance the contrast, the basic idea is to increase the dynamic range of pixel values so that the pixels get easy to distinguish. Point-wise transformations are used that operate on each pixel independently to modify its gray level. A point-wise transformation $g(x)$ operates on the input image A to produce a new image B via:

$$B(i, j) = g(A(i, j)), i = 0, \dots, N - 1; j = 0, \dots, M - 1 \quad (2)$$

where, $N \times M$ is the size of the image.

The transformation $g(x)$ should satisfy two criteria i.e. $g(x)$ is monotonically increasing and it has the same range as x . Such transformations could be characterized as having either a fixed functional form or an adaptive form in which case it is

determined from the histogram of the original image and the desired one. Some non-linear stretching function of fixed form are logarithmic, exponential, and power law transformation. Power law is used by devices for acquiring images. The exponent of power law is referred to as *gamma* by convention and therefore this transformation is also called as *gamma correction*. The general form is given by,

$$g(f) = cf^\gamma \quad (3)$$

For the detection of editing operations, existing forensic detectors work on the assumption that although the applied operation may not leave any visual clue in the image but it does disturb the statistics of the image. Stamm and Liu [14] proposed a method for detection of contrast enhancement based on the statistical intrinsic fingerprints left by contrast enhancement on the image's histogram. When applied to an image it causes an increase in energy in the image's histogram as shown in Fig. 1. In Fig. 1a, Lena's unaltered picture is shown along with its histogram which has an almost smooth envelop. However, Fig. 1b shows the altered version along with the histogram which has quite sudden peaks and gaps in it. Such a change in the histogram causes an increase in energy in its Discrete Fourier Transform (DFT). This increase in energy

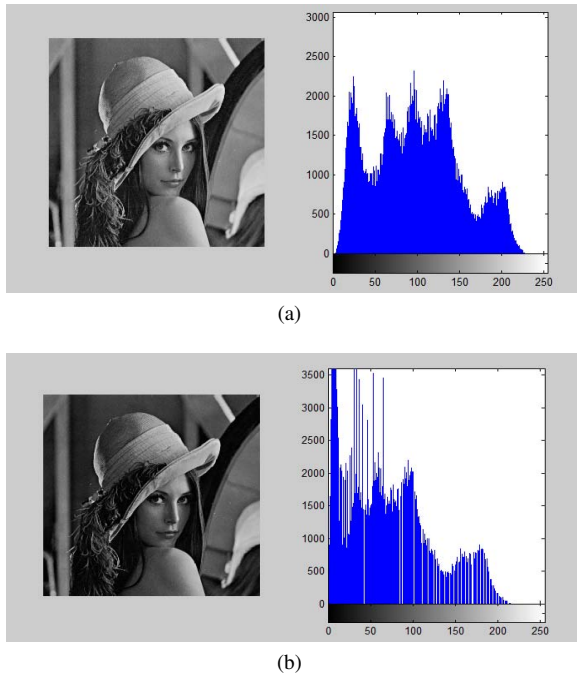


Fig. 1: (a) Original images and its histogram and (b) Contrast changed image and its histogram

spreads across the higher frequencies and can be measured from the magnitude of DFT. Their detector performs well if the images are not JPEG compressed. Lin et al.[15] exploited acquisition fingerprints as Color Filter Arrays (CFAs) are used to detect the color component of the incoming light. Since, only one component could be detected for each pixel position, rest of the components are interpolated using color interpolation algorithms which introduce inter-channel similarity between the high frequency components. When contrast enhancement is applied to an image this inter-channel similarity gets disturbed.

So, they proposed a matrix S as the measure of Inter-channel high-frequency similarity which can be computed to detect contrast enhancement. In another research paper, Cao et al. [16] exploited the presence of gap bins in the histogram of contrast enhanced images and their absence in unaltered uncompressed as well as compressed image's histogram to detect contrast enhancement.

III. PROPOSED METHOD

Benford's law has already been used in digital image forensics for the detection of various forgeries as discussed in Section I. In this work, focus is on the images that are first compressed before applying contrast enhancement operation since this helps in destroying the characteristic fingerprints of contrast enhancement operation. For such images, the Benford's law based features are extracted and analysed using the system model as shown in Fig. 2. As shown in the

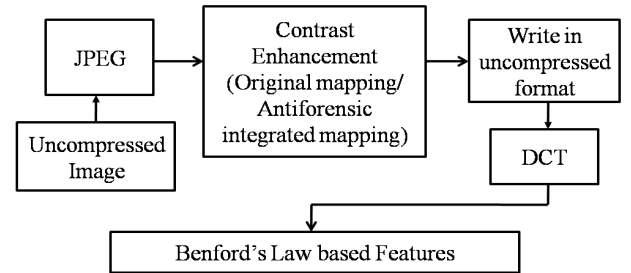


Fig. 2: System model of the proposed idea

system model, while applying contrast enhancement two kind of mappings are used. One is the general contrast enhancement mapping commonly known as gamma correction (Equation. 3) and the other is an anti-forensic scheme integrated mapping as proposed in [17]. In the following Fig. 3, Fig. 3a and Fig. 3b shows the fitting of significant digits of DCT coefficients for an unaltered image and its altered version using general contrast enhancement mapping taken from the UCID dataset[18], respectively. In the latter case, although the distribution of most of the digits look similar to Benford's distribution but a quite deviation is observed at digit 7. Next, we measured the quality of fitting using Chi-square divergence[19]. The χ^2 divergence is 0.0051 for unaltered image and 0.0791 for altered image which shows a change in the distribution of first digits of DCT coefficients. For the purpose of concrete investigation, we calculated the χ^2 divergences for all the 1338 images of the UCID dataset altered using different values of gamma. The mean of χ^2 divergences for the fitting of altered UCID images is 0.0180 which is quite greater than 0.0034, mean of χ^2 divergences for unaltered UCID images as observed in [9]. These results indicate a possibility of using Benford's law based features for the detection of contrast enhancement.

Next, an experiment is conducted on images compressed with three different quality factors i.e 90, 70, and 50. For creating the sets of altered images, we use gamma correction mapping with four different values of gamma i.e. 0.5, 0.9, 1.5 and 2.0. The Benford's law based features are extracted corresponding to each quality factor and gamma value. As shown in the Table I, the mean chi-square divergences are recorded for each pair of parameters. It is interesting to note

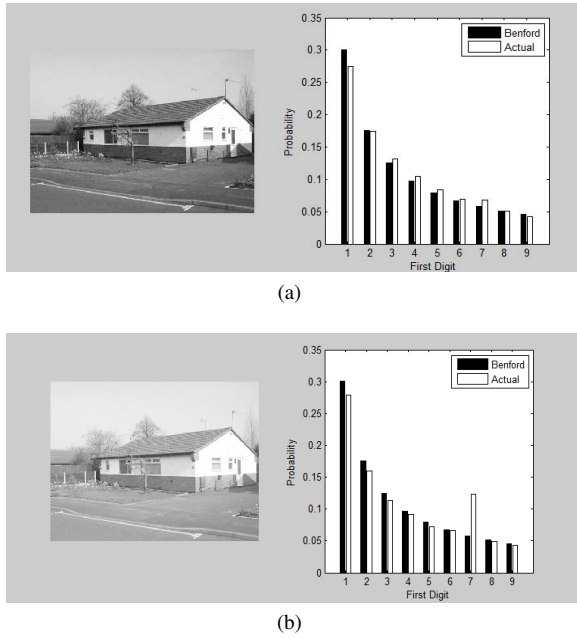


Fig. 3: (a)Original image and (b)Altered image along with their DCT coefficient's significant digit's fitting to Benford's law

TABLE I: Mean Chi-square divergences

Quality Factor	Unaltered	Altered			
		$\gamma = 0.5$	$\gamma = 0.9$	$\gamma = 1.5$	$\gamma = 2.0$
90	0.0112	0.0665	0.0107	0.0081	0.0061
70	0.0109	0.0634	0.0091	0.0076	0.0063
50	0.0126	0.0667	0.0092	0.0057	0.0051

that as the value of gamma increases, the divergence of DCT coefficients' significant digits' frequencies from the Benford's distribution tends to reduce. This pattern could further be exploited to detect the value of gamma used for altering the contrast of the image. This observation has an interesting second side as well that in order to make the DCT significant digit features follow the Benford's law, one can use higher values of gamma.

IV. RESULTS

A. Original Contrast Enhancement Detection

For the purpose of large scale testing, we used UCID (Uncompressed Color Image Database)[18] which consists of 1338 images. To make a set of unaltered grayscale images, we took the green color layer of each of the images from the dataset. The unaltered grayscale images were compressed using JPEG coding with three different quality factors (50, 70, and 90) for the purpose of creating a set of compressed grayscale images. The Gamma correction mapping was applied(Equation.2) to the set of unaltered grayscale images using γ values in the range 0.5 to 2.0 to create a set of contrast enhanced images. For the Benford's law based features, attempt was made to train an SVM classifier using 10-fold cross validation and the resultant models' accuracies are listed in the Tables II. For each model, table shows the detection probability P_d and

false alarm rate P_{fa} . These percentages represent the number of altered images correctly classified as altered and the number of unaltered images incorrectly classified as altered, respectively. The extracted features are showing comparable performance with respect to the results reported in [16] where the authors achieved 100% detection accuracy while the Stamm and Liu's [14] approach becomes random guess at lower quality factors.

TABLE II: Detection performance of Benford's features for original contrast enhancement mapping

Gamma	QF=50		QF=70		QF=90	
	P_d	P_{fa}	P_d	P_{fa}	P_d	P_{fa}
0.5	96.8%	4.8%	98.7%	4%	98.1%	4.1%
0.9	93.7%	7.3%	94.8%	6.1%	93.1%	6.6%
1.5	97.6%	4.2%	96.7%	3.5%	92.7%	6.3%
2.0	97.7%	4.3%	97.2%	5.1%	94%	5.6%

B. Anti-forensic Scheme Integrated Contrast Enhancement Detection

In the research paper [17], the authors proposed an integrated attack on Contrast enhancement detection forensic technique using *Local Random Dithering*. It is interesting to note that with their proposed integrated mapping, the resultant contrast enhanced image is comparable in quality with images contrast enhanced using original mapping given in Equation 3. This can be clearly seen in the example image shown in Fig. 4.

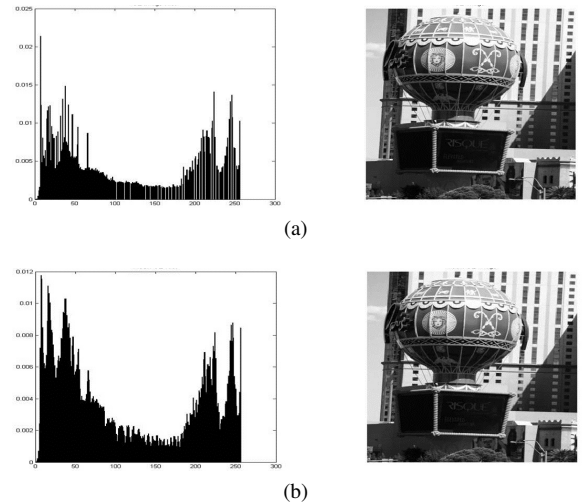


Fig. 4: Histograms along with corresponding images contrast enhanced using (a)Original mapping, and (b)Anti-forensic attack integrated mapping

However, the histogram of the image modified using anti-forensic attack integrated mapping has no zero-height gap bins. The Cao et al. [16] contrast enhancement detection technique is based on the presence of such bins. Histograms of randomly selected 100 images from the UCID dataset with each image compressed with three different quality factors (50, 70, and 90) are analysed to detect the presence of gap bins. Three cases are evaluated i.e unaltered images, images altered using general mapping and images altered using anti-forensic integrated

TABLE III: Analysis of Cao et al. [16] showing average number of zero-height gap bins

	Gamma	JPEG QF		
		50	70	90
Unaltered	(0.5-2.0)	0	0	0
Original Contrast Enhancement Mapping	0.5	15.57	16.85	15.97
	0.9	5.83	6.17	5.79
	1.5	23.02	26.78	24.27
	2.0	31.20	36.84	33.33
Anti-forensic integrated Mapping	(0.5-2.0)	0	0	0

mapping with different values of gamma. Table III shows the mean number of gap bins detected in each case.

As evident from Table III, anti-forensic integrated mapping does not leave gap bins of zero height in the resultant image's histogram. Hence, their proposed technique cannot detect contrast enhancement in case of anti-forensic attack.

For testing the applicability of Benford's law based features in this scenario, the set of altered images is created using anti-forensic integrated mapping of contrast enhancement along with different values of gamma. Again, the SVM classifiers are trained using 10 fold cross validation. Below Table IV shows the performance of Benford features in the detection of anti-forensic integrated contrast enhancement mapping. The detection accuracies in-fact improved in this case particularly for the value of gamma=0.9.

TABLE IV: Detection performance of Benford's features for anti-forensic integrated contrast enhancement mapping

Gamma	QF=50		QF=70		QF=90	
	P_d	P_{fa}	P_d	P_{fa}	P_d	P_{fa}
0.5	97.8%	1.5%	97%	1%	97%	2%
0.9	99%	1.5%	99.2%	1.8%	98%	1.7%
1.5	99.1%	1.4%	99.1%	4%	98.5%	1.6%
2.0	99.2%	1.3%	98%	2.5%	99%	1.9%

V. CONCLUSION

In this paper, the possibility of using Benford's Law for the detection of contrast enhancement is analysed. A pattern is observed based on the mean divergences of the extracted features from the Benford's law distribution. This pattern cannot be used in the first place to estimate the value of gamma, however further analysis in this direction could give a way to estimate the value of gamma used to alter the image. Also, Benford's law based features show promising performance in the detection of contrast enhancement in case of both the original contrast enhancement mapping and the anti-forensic attack integrated mapping. A comparison has been made with an existing approach[16] which clearly fails in case of an anti-forensic attack while Benford's law based features show good robustness to this attack. The performance can be further improved by either reducing the size of the feature space by identifying the more influencing features or by introducing a new promising feature.

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